

The Notebooks • Era I — Count

Stone tablets / tally notches

Testing Mode • UK • IB • US

Exercises

T.I.1.001 • Count in Roman numerals to XII • core • UK_KS2

- (a) [never/neutral/1mk] Convert 4 to Roman numerals (skill: Count in Roman numerals to XII).
- (b) [never/UK/1mk] What Arabic number is I?
- (c) [never/UK/2mk] A page is numbered IV. Write the next page number in Roman numerals.

T.I.1.002 • Cardinality: last number said is how many • core • US_CCSS_K5,UK_KS2

- (a) [never/neutral/1mk] Continue (step 2): 4, 6, 8, 10, ... Write the next two terms.
- (b) [never/US/1mk] Fill ■: 4, 6, 8, 10, ■, 14, 16.
- (c) [never/US/2mk] Count back from 16 in steps of 2. List the first four numbers said.

T.I.1.003 • Subitize small sets without counting • core • US_CCSS_K5,UK_KS2

- (a) [never/neutral/1mk] Continue (step 2): 2, 4, 6, 8, ... Write the next two terms.
- (b) [never/US/1mk] Fill ■: 2, 4, 6, 8, ■, 12, 14.
- (c) [never/US/2mk] Count back from 14 in steps of 2. List the first four numbers said.

T.I.1.004 • Skip-count by 2, 5, 10 exam drills • core • UK_KS2,US_CCSS_K5

- (a) [never/neutral/1mk] Continue (step 10): 40, 50, 60, 70, ... Write the next two terms.
- (b) [never/UK/1mk] Fill ■: 40, 50, 60, ■, 80, 90, 100.
- (c) [never/UK/2mk] Count back from 100 in steps of 10. List the first four numbers said.

T.I.1.005 • Count in steps of 3 and 4 • core • UK_KS2

- (a) [never/neutral/1mk] Continue (step 3): 9, 12, 15, 18, ... Write the next two terms.
- (b) [never/UK/1mk] Fill ■: 9, 12, ■, 18, 21, 24, 27.
- (c) [never/UK/2mk] Count back from 27 in steps of 3. List the first four numbers said.

T.I.1.006 • Count back through zero into negatives • challenge • UK_KS2,UK_KS3

- (a) [never/neutral/1mk] Continue (step 2): 4, 6, 8, 10, ... Write the next two terms.
- (b) [never/UK/1mk] Fill ■: 4, 6, 8, 10, ■, 14, 16.
- (c) [never/UK/2mk] Count back from 16 in steps of 2. List the first four numbers said.

T.I.2.001 • Compare using more/fewer/equal SATs stems • core • UK_KS2

- (a) [never/neutral/1mk] Partition 853 into hundreds, tens and ones.
- (b) [never/UK/1mk] Write 853 in expanded form.
- (c) [never/UK/2mk] Exchange: 15 ones equal how many tens and ones?

T.I.2.002 • Order lengths and masses with units • core • UK_KS2,US_CCSS_K5

- (a) [never/neutral/1mk] Partition 342 into hundreds, tens and ones.
- (b) [never/UK/1mk] Write 342 in expanded form.
- (c) [never/UK/2mk] Exchange: 14 ones equal how many tens and ones?

T.I.2.003 • Order numbers to 100 with inequality signs • core • UK_KS2,US_CCSS_K5

- (a) [never/neutral/1mk] Partition 256 into hundreds, tens and ones.
- (b) [never/UK/1mk] Write 256 in expanded form.
- (c) [never/UK/2mk] Exchange: 15 ones equal how many tens and ones?

T.I.3.001 • Partition with UK part-whole models • core • UK_KS2

- (a) [never/neutral/1mk] Partition 391 into hundreds, tens and ones.
- (b) [never/UK/1mk] Write 391 in expanded form.
- (c) [never/UK/2mk] Exchange: 19 ones equal how many tens and ones?

T.I.3.002 · Write numbers in expanded form (UK wording) · core · UK_KS2

- (a) [never/neutral/1mk] Partition 857 into hundreds, tens and ones.
- (b) [never/UK/1mk] Write 857 in expanded form.
- (c) [never/UK/2mk] Exchange: 15 ones equal how many tens and ones?

T.I.3.003 · Roman numerals to 100 (I–C) · core · UK_KS2

- (a) [never/neutral/1mk] Convert 74 to Roman numerals (skill: Roman numerals to 100 (I–C)).
- (b) [never/UK/1mk] What Arabic number is X?
- (c) [never/UK/2mk] A page is numbered LXXIV. Write the next page number in Roman numerals.

T.I.3.004 · Roman numerals to 1,000 (M) · challenge · UK_KS2

- (a) [never/neutral/1mk] Convert 500 to Roman numerals (skill: Roman numerals to 1,000 (M)).
- (b) [never/UK/1mk] What Arabic number is XL?
- (c) [never/UK/2mk] A page is numbered D. Write the next page number in Roman numerals.

T.I.3.005 · Exchange ten ones for one ten · core · UK_KS2,US_CCSS_K5

- (a) [never/neutral/1mk] Partition 189 into hundreds, tens and ones.
- (b) [never/UK/1mk] Write 189 in expanded form.
- (c) [never/UK/2mk] Exchange: 18 ones equal how many tens and ones?

T.I.3.006 · Flexibly partition numbers many ways · core · UK_KS2,US_CCSS_K5

- (a) [never/neutral/1mk] Partition 928 into hundreds, tens and ones.
- (b) [never/UK/1mk] Write 928 in expanded form.
- (c) [never/UK/2mk] Exchange: 12 ones equal how many tens and ones?

T.I.4.001 · Column addition (Appendix formal method) · core · UK_KS2

- (a) [never/neutral/2mk] Column addition: $707 + 346$.
- (b) [never/UK/2mk] Work out $717 + 341$.
- (c) [never/UK/2mk] Score 707 then 346 more. Total?

T.I.4.002 · Column subtraction with exchange · core · UK_KS2

- (a) [never/neutral/2mk] Column subtraction: $514 - 281$.
- (b) [never/UK/2mk] Calculate $511 - 283$.
- (c) [never/UK/2mk] A library has 514 books; 281 are loaned. How many remain?

T.I.4.003 · Number bonds to 10 and 20 speed packs · core · UK_KS2

- (a) [never/neutral/2mk] Column addition: $277 + 628$.
- (b) [never/UK/2mk] Work out $287 + 623$.
- (c) [never/UK/2mk] Score 277 then 628 more. Total?

T.I.4.004 · Add/subtract money in £ and p · core · UK_KS2

- (a) [never/neutral/2mk] Column subtraction: $623 - 348$.
- (b) [never/UK/2mk] Calculate $620 - 350$.
- (c) [never/UK/2mk] A library has 623 books; 348 are loaned. How many remain?

T.I.4.005 · Worded multi-step add/sub (SATs style) · challenge · UK_KS2

- (a) [never/neutral/2mk] Column subtraction: $986 - 726$.
- (b) [never/UK/2mk] Calculate $983 - 728$.
- (c) [never/UK/2mk] A library has 986 books; 726 are loaned. How many remain?

T.I.4.006 · Near doubles addition strategies · core · UK_KS2,US_CCSS_K5

- (a) [never/neutral/2mk] Column addition: $656 + 254$.
- (b) [never/UK/2mk] Work out $666 + 249$.
- (c) [never/UK/2mk] Score 656 then 254 more. Total?

T.I.4.007 · Bridging through 10 subtraction · core · UK_KS2

- (a) [never/neutral/2mk] Column subtraction: $521 - 420$.
- (b) [never/UK/2mk] Calculate $518 - 422$.
- (c) [never/UK/2mk] A library has 521 books; 420 are loaned. How many remain?

T.I.4.008 · Inverse operations check add/sub · core · UK_KS2,US_CCSS_K5

- (a) [never/neutral/2mk] Column subtraction: $808 - 358$.
- (b) [never/UK/2mk] Calculate $805 - 360$.
- (c) [never/UK/2mk] A library has 808 books; 358 are loaned. How many remain?

T.I.4.009 · Column methods with money £.p · core · UK_KS2

- (a) [never/neutral/2mk] Column addition: $350 + 552$.
- (b) [never/UK/2mk] Work out $360 + 547$.
- (c) [never/UK/2mk] Score 350 then 552 more. Total?

T.I.5.001 · Repeating pattern SAT stems · core · UK_KS2,US_CCSS_K5

- (a) [never/neutral/1mk] Continue (step 2): 2, 4, 6, 8, ... Write the next two terms.
- (b) [never/UK/1mk] Fill ■: 2, 4, 6, ■, 10, 12, 14.
- (c) [never/UK/2mk] Count back from 14 in steps of 2. List the first four numbers said.

T.I.5.002 · Missing-number sentences with boxes · core · UK_KS2

- (a) [never/neutral/1mk] Continue (step 2): 6, 8, 10, 12, ... Write the next two terms.
- (b) [never/UK/1mk] Fill ■: 6, 8, 10, 12, ■, 16, 18.
- (c) [never/UK/2mk] Count back from 18 in steps of 2. List the first four numbers said.

T.I.6.001 · Name 2D shapes with UK vocabulary · core · UK_KS2

- (a) [never/neutral/1mk] [Name 2D shapes with UK vocabulary] Which view is the plan — front, side, or top?
- (b) [never/UK/1mk] Cuboid $8 \times 4 \times 2$: how many rectangles in a correct net?
- (c) [allowed/UK/2mk] Describe the front elevation of a square-based pyramid (base 8) standing on its base.

T.I.6.002 · Turns: quarter, half, three-quarter, full · core · UK_KS2

- (a) [never/neutral/1mk] [Turns: quarter, half, three-quarter, full] Which view is the plan — front, side, or top?
- (b) [never/UK/1mk] Cuboid $5 \times 6 \times 4$: how many rectangles in a correct net?
- (c) [allowed/UK/2mk] Describe the front elevation of a square-based pyramid (base 5) standing on its base.

T.I.6.003 · Left/right and compass directions early · core · UK_KS2

- (a) [never/neutral/1mk] [Left/right and compass directions early] Which view is the plan — front, side, or top?
- (b) [never/UK/1mk] Cuboid $9 \times 4 \times 4$: how many rectangles in a correct net?
- (c) [allowed/UK/2mk] Describe the front elevation of a square-based pyramid (base 9) standing on its base.

T.I.6.004 · Symmetry: complete mirror patterns · core · UK_KS2,US_CCSS_K5

- (a) [never/neutral/1mk] [Symmetry: complete mirror patterns] Which view is the plan — front, side, or top?
- (b) [never/UK/1mk] Cuboid $5 \times 7 \times 5$: how many rectangles in a correct net?
- (c) [allowed/UK/2mk] Describe the front elevation of a square-based pyramid (base 5) standing on its base.

T.I.7.001 · Tell time to the nearest minute (analogue) · core · UK_KS2

- (a) [never/neutral/1mk] Write 7:25 pm in 24-hour time.
- (b) [never/UK/2mk] Start 7:25; duration 95 min. End time?
- (c) [never/UK/2mk] Train 14:25 → 16:00. Journey length?

T.I.7.002 · 12-hour clock with am/pm · core · UK_KS2

- (a) [never/neutral/1mk] Write 11:35 pm in 24-hour time.
- (b) [never/UK/2mk] Start 11:35; duration 35 min. End time?
- (c) [never/UK/2mk] Train 14:35 → 15:10. Journey length?

T.I.7.003 · 24-hour clock conversion · core · UK_KS2

- (a) [never/neutral/1mk] Write 1:00 pm in 24-hour time.
- (b) [never/UK/2mk] Start 1:00; duration 95 min. End time?
- (c) [never/UK/2mk] Train 14:00 → 15:35. Journey length?

T.I.7.004 · Read simple bus/train timetables · challenge · UK_KS2

- (a) [never/neutral/1mk] Write 6:00 pm in 24-hour time.
- (b) [never/UK/2mk] Start 6:00; duration 70 min. End time?
- (c) [never/UK/2mk] Train 14:00 → 15:10. Journey length?

T.I.7.005 · British coins and notes identification · core · UK_KS2

- (a) [never/neutral/1mk] Total of £0.92 and £0.48?
- (b) [never/UK/2mk] Pay £5.00 for goods costing £1.40. Change?
- (c) [never/UK/3mk] Buy three items £0.92, £0.48, £2.16 with a £10 note. Enough? Change?

T.I.7.006 · Give change from £1, £5, £10, £20 · core · UK_KS2

- (a) [never/neutral/1mk] Total of £2.48 and £0.68?
- (b) [never/UK/2mk] Pay £5.00 for goods costing £3.16. Change?
- (c) [never/UK/3mk] Buy three items £2.48, £0.68, £1.63 with a £10 note. Enough? Change?

T.I.7.007 · Temperature in °C: read and compare · core · UK_KS2

- (a) [never/neutral/1mk] Convert 5 cm to mm.
- (b) [never/UK/2mk] Which is heavier: 500 g or 2 kg? (compare in g)
- (c) [never/UK/1mk] A jug holds 500 ml. How many litres?

T.I.7.008 · Tally charts exam pack · core · UK_KS2

- (a) [allowed/neutral/1mk] Mean of 6, 11, 11, 11, 14, 18?
- (b) [allowed/UK/1mk] Range of 6, 11, 11, 11, 14, 18?
- (c) [allowed/UK/2mk] Mean of five numbers is 13. Four are 6, 11, 11, 11. Find the fifth.

T.I.7.009 · Pictograms with keys (one symbol = many) · core · UK_KS2

- (a) [allowed/neutral/1mk] Mean of 4, 6, 9, 12, 13, 18?
- (b) [allowed/UK/1mk] Range of 4, 6, 9, 12, 13, 18?
- (c) [allowed/UK/2mk] Mean of five numbers is 11. Four are 4, 6, 9, 12. Find the fifth.

T.I.7.010 · Block diagrams / bar charts simple scale · core · UK_KS2

- (a) [allowed/neutral/1mk] Mean of 3, 8, 9, 12, 13, 18?
- (b) [allowed/UK/1mk] Range of 3, 8, 9, 12, 13, 18?
- (c) [allowed/UK/2mk] Mean of five numbers is 11. Four are 3, 8, 9, 12. Find the fifth.

T.I.7.011 · Measure length in cm and mm · core · UK_KS2

- (a) [never/neutral/1mk] Convert 16 cm to mm.
- (b) [never/UK/2mk] Which is heavier: 1600 g or 6 kg? (compare in g)
- (c) [never/UK/1mk] A jug holds 1600 ml. How many litres?

T.I.7.012 · Mass in g and kg comparison · core · UK_KS2

- (a) [never/neutral/1mk] Convert 16 cm to mm.
- (b) [never/UK/2mk] Which is heavier: 1600 g or 7 kg? (compare in g)
- (c) [never/UK/1mk] A jug holds 1600 ml. How many litres?

T.I.7.013 · Capacity in ml and l · core · UK_KS2

- (a) [never/neutral/1mk] Convert 3 cm to mm.
- (b) [never/UK/2mk] Which is heavier: 300 g or 8 kg? (compare in g)
- (c) [never/UK/1mk] A jug holds 300 ml. How many litres?

T.I.7.014 · CCSS line plots with fractional lengths · challenge · US_CCSS_K5,UK_KS2

- (a) [allowed/neutral/1mk] Mean of 3, 7, 8, 9, 12, 14?
- (b) [allowed/US/1mk] Range of 3, 7, 8, 9, 12, 14?
- (c) [allowed/US/2mk] Mean of five numbers is 8. Four are 3, 7, 8, 9. Find the fifth.

T.I.7.015 · Elapsed time across hour boundaries · challenge · UK_KS2

- (a) [never/neutral/1mk] Write 1:25 pm in 24-hour time.
- (b) [never/UK/2mk] Start 1:25; duration 95 min. End time?
- (c) [never/UK/2mk] Train 14:25 → 16:00. Journey length?

T.I.7.016 · Calendar: days in months fluency · core · UK_KS2

- (a) [never/neutral/1mk] Write 7:05 pm in 24-hour time.
- (b) [never/UK/2mk] Start 7:05; duration 25 min. End time?
- (c) [never/UK/2mk] Train 14:05 → 14:30. Journey length?

T.I.7.017 · Compare durations in hours and minutes · core · UK_KS2

- (a) [never/neutral/1mk] Write 10:50 pm in 24-hour time.
- (b) [never/UK/2mk] Start 10:50; duration 95 min. End time?
- (c) [never/UK/2mk] Train 14:50 → 16:25. Journey length?

T.I.7.018 · Weigh using scales reading dials · core · UK_KS2

- (a) [never/neutral/1mk] Convert 16 cm to mm.
- (b) [never/UK/2mk] Which is heavier: 1600 g or 15 kg? (compare in g)
- (c) [never/UK/1mk] A jug holds 1600 ml. How many litres?

T.I.7.019 · Capacity practical measuring jugs · core · UK_KS2

- (a) [never/neutral/1mk] Convert 10 cm to mm.
- (b) [never/UK/2mk] Which is heavier: 1000 g or 9 kg? (compare in g)
- (c) [never/UK/1mk] A jug holds 1000 ml. How many litres?

T.I.7.020 · Money change multi-item shop 1 · core · UK_KS2

- (a) [never/neutral/1mk] Total of £0.66 and £1.43?
- (b) [never/UK/2mk] Pay £5.00 for goods costing £2.09. Change?
- (c) [never/UK/3mk] Buy three items £0.66, £1.43, £2.59 with a £10 note. Enough? Change?

T.I.7.021 · Money change multi-item shop 2 · core · UK_KS2

- (a) [never/neutral/1mk] Total of £1.07 and £1.55?
- (b) [never/UK/2mk] Pay £5.00 for goods costing £2.62. Change?
- (c) [never/UK/3mk] Buy three items £1.07, £1.55, £1.84 with a £10 note. Enough? Change?

T.I.7.022 · Money change multi-item shop 3 · core · UK_KS2

- (a) [never/neutral/1mk] Total of £0.50 and £2.25?
- (b) [never/UK/2mk] Pay £5.00 for goods costing £2.75. Change?
- (c) [never/UK/3mk] Buy three items £0.50, £2.25, £2.98 with a £10 note. Enough? Change?

T.I.7.023 · Money change multi-item shop 4 · core · UK_KS2

- (a) [never/neutral/1mk] Total of £1.37 and £0.55?
- (b) [never/UK/2mk] Pay £5.00 for goods costing £1.92. Change?
- (c) [never/UK/3mk] Buy three items £1.37, £0.55, £1.53 with a £10 note. Enough? Change?

T.I.7.024 · Timetable reading scenario 1 · core · UK_KS2

- (a) [never/neutral/1mk] Write 10:50 pm in 24-hour time.
- (b) [never/UK/2mk] Start 10:50; duration 95 min. End time?
- (c) [never/UK/2mk] Train 14:50 → 16:25. Journey length?

T.I.7.025 · Timetable reading scenario 2 · core · UK_KS2

- (a) [never/neutral/1mk] Write 7:55 pm in 24-hour time.
- (b) [never/UK/2mk] Start 7:55; duration 95 min. End time?
- (c) [never/UK/2mk] Train 14:55 → 16:30. Journey length?

T.I.7.026 · Timetable reading scenario 3 · core · UK_KS2

- (a) [never/neutral/1mk] Write 10:30 pm in 24-hour time.
- (b) [never/UK/2mk] Start 10:30; duration 95 min. End time?
- (c) [never/UK/2mk] Train 14:30 → 16:05. Journey length?

Answer Key — Era I

Brief solutions. Full reasoning in machine JSON.

T.I.1.001

- (a) IV — *Standard Roman mapping* → IV.
- (b) $1 - I = 1$.
- (c) $V - 4 + 1 = 5 \rightarrow V$.

T.I.1.002

- (a) 12, 14 — *Add 2 each time*.
- (b) 12 — *Missing term* = 12.
- (c) 16, 14, 12, 10 — *Subtract the step repeatedly*.

T.I.1.003

- (a) 10, 12 — *Add 2 each time*.
- (b) 10 — *Missing term* = 10.
- (c) 14, 12, 10, 8 — *Subtract the step repeatedly*.

T.I.1.004

- (a) 80, 90 — *Add 10 each time*.
- (b) 70 — *Missing term* = 70.
- (c) 100, 90, 80, 70 — *Subtract the step repeatedly*.

T.I.1.005

- (a) 21, 24 — *Add 3 each time*.
- (b) 15 — *Missing term* = 15.
- (c) 27, 24, 21, 18 — *Subtract the step repeatedly*.

T.I.1.006

- (a) 12, 14 — *Add 2 each time*.
- (b) 12 — *Missing term* = 12.
- (c) 16, 14, 12, 10 — *Subtract the step repeatedly*.

T.I.2.001

- (a) 8 hundreds + 5 tens + 3 ones — *Read each digit's place*.
- (b) $800 + 50 + 3$ — *Expanded notation*.
- (c) 1 ten and 5 ones (from 10 ones) plus leftover context: 15 ones = 1 tens + 5 ones — *10 ones = 1 ten*.

T.I.2.002

- (a) 3 hundreds + 4 tens + 2 ones — *Read each digit's place*.
- (b) $300 + 40 + 2$ — *Expanded notation*.
- (c) 1 ten and 4 ones (from 10 ones) plus leftover context: 14 ones = 1 tens + 4 ones — *10 ones = 1 ten*.

T.I.2.003

- (a) 2 hundreds + 5 tens + 6 ones — *Read each digit's place*.
- (b) $200 + 50 + 6$ — *Expanded notation*.
- (c) 1 ten and 5 ones (from 10 ones) plus leftover context: 15 ones = 1 tens + 5 ones — *10 ones = 1 ten*.

T.I.3.001

- (a) 3 hundreds + 9 tens + 1 ones — *Read each digit's place*.
- (b) $300 + 90 + 1$ — *Expanded notation*.
- (c) 1 ten and 9 ones (from 10 ones) plus leftover context: 19 ones = 1 tens + 9 ones — *10 ones = 1 ten*.

T.I.3.002

- (a) 8 hundreds + 5 tens + 7 ones — *Read each digit's place*.
- (b) $800 + 50 + 7$ — *Expanded notation*.
- (c) 1 ten and 5 ones (from 10 ones) plus leftover context: 15 ones = 1 tens + 5 ones — *10 ones = 1 ten*.

T.I.3.003

- (a) LXXIV — *Standard Roman mapping* → LXXIV.
- (b) 10 — $X = 10$.
- (c) LXXV — $74 + 1 = 75 \rightarrow LXXV$.

T.I.3.004

- (a) D — *Standard Roman mapping* → D.
- (b) 40 — $XL = 40$.
- (c) DI — $500 + 1 = 501 \rightarrow DI$.

T.I.3.005

- (a) 1 hundreds + 8 tens + 9 ones — *Read each digit's place*.

- (b) $100 + 80 + 9$ — *Expanded notation.*
(c) 1 ten and 8 ones (from 10 ones) plus leftover context: $18 \text{ ones} = 1 \text{ tens} + 8 \text{ ones}$ — *10 ones = 1 ten.*

T.I.3.006

- (a) 9 hundreds + 2 tens + 8 ones — *Read each digit's place.*
(b) $900 + 20 + 8$ — *Expanded notation.*
(c) 1 ten and 2 ones (from 10 ones) plus leftover context: $12 \text{ ones} = 1 \text{ tens} + 2 \text{ ones}$ — *10 ones = 1 ten.*

T.I.4.001

- (a) 1053 — *Add with carries.*
(b) 1058 — *Column addition.*
(c) $1053 - 707 = 346$.

T.I.4.002

- (a) 233 — *Exchange as needed.*
(b) 228 — *Column method.*
(c) $233 - 514 = -281$.

T.I.4.003

- (a) 905 — *Add with carries.*
(b) 910 — *Column addition.*
(c) $905 - 277 = 628$.

T.I.4.004

- (a) 275 — *Exchange as needed.*
(b) 270 — *Column method.*
(c) $275 - 623 = 348$.

T.I.4.005

- (a) 260 — *Exchange as needed.*
(b) 255 — *Column method.*
(c) $260 - 986 = -726$.

T.I.4.006

- (a) 910 — *Add with carries.*
(b) 915 — *Column addition.*
(c) $910 - 656 = 254$.

T.I.4.007

- (a) 101 — *Exchange as needed.*
(b) 96 — *Column method.*
(c) $101 - 521 = -420$.

T.I.4.008

- (a) 450 — *Exchange as needed.*
(b) 445 — *Column method.*
(c) $450 - 808 = -358$.

T.I.4.009

- (a) 902 — *Add with carries.*
(b) 907 — *Column addition.*
(c) $902 - 350 = 552$.

T.I.5.001

- (a) 10, 12 — *Add 2 each time.*
(b) 8 — *Missing term = 8.*
(c) 14, 12, 10, 8 — *Subtract the step repeatedly.*

T.I.5.002

- (a) 14, 16 — *Add 2 each time.*
(b) 14 — *Missing term = 14.*
(c) 18, 16, 14, 12 — *Subtract the step repeatedly.*

T.I.6.001

- (a) Top view (from above) — *Plan = bird's-eye.*
(b) 6 — *Six faces.*
(c) Isosceles triangle with base L — *Front sees triangular profile.*

T.I.6.002

- (a) Top view (from above) — *Plan = bird's-eye.*
(b) 6 — *Six faces.*
(c) Isosceles triangle with base L — *Front sees triangular profile.*

T.I.6.003

- (a) Top view (from above) — *Plan = bird's-eye.*
- (b) 6 — *Six faces.*
- (c) Isosceles triangle with base L — *Front sees triangular profile.*

T.I.6.004

- (a) Top view (from above) — *Plan = bird's-eye.*
- (b) 6 — *Six faces.*
- (c) Isosceles triangle with base L — *Front sees triangular profile.*

T.I.7.001

- (a) 19:25 — *Add 12 for pm (except 12 pm).*
- (b) 9:00 — *Add minutes; roll hours.*
- (c) 95 minutes — *Arrival – departure.*

T.I.7.002

- (a) 23:35 — *Add 12 for pm (except 12 pm).*
- (b) 12:10 — *Add minutes; roll hours.*
- (c) 35 minutes — *Arrival – departure.*

T.I.7.003

- (a) 13:00 — *Add 12 for pm (except 12 pm).*
- (b) 2:35 — *Add minutes; roll hours.*
- (c) 95 minutes — *Arrival – departure.*

T.I.7.004

- (a) 18:00 — *Add 12 for pm (except 12 pm).*
- (b) 7:10 — *Add minutes; roll hours.*
- (c) 70 minutes — *Arrival – departure.*

T.I.7.005

- (a) £1.40 — *Add pence; convert.*
- (b) £3.60 — *Subtract.*
- (c) Yes, £6.44 — *Compare sum with 1000p.*

T.I.7.006

- (a) £3.16 — *Add pence; convert.*
- (b) £1.84 — *Subtract.*
- (c) Yes, £5.21 — *Compare sum with 1000p.*

T.I.7.007

- (a) 50 mm — $\times 10$.
- (b) 2 kg — *Convert to same unit.*
- (c) 0.5 l — $\div 1000$.

T.I.7.008

- (a) 11.83 — *Sum $71 \div 6$.*
- (b) 12 — *max–min.*
- (c) 26 — $5 \times \text{mean} - \text{sum of four}$.

T.I.7.009

- (a) 10.33 — *Sum $62 \div 6$.*
- (b) 14 — *max–min.*
- (c) 24 — $5 \times \text{mean} - \text{sum of four}$.

T.I.7.010

- (a) 10.5 — *Sum $63 \div 6$.*
- (b) 15 — *max–min.*
- (c) 23 — $5 \times \text{mean} - \text{sum of four}$.

T.I.7.011

- (a) 160 mm — $\times 10$.
- (b) 6 kg — *Convert to same unit.*
- (c) 1.6 l — $\div 1000$.

T.I.7.012

- (a) 160 mm — $\times 10$.
- (b) 7 kg — *Convert to same unit.*
- (c) 1.6 l — $\div 1000$.

T.I.7.013

- (a) 30 mm — $\times 10$.
(b) 8 kg — *Convert to same unit.*
(c) 0.3 l — $\div 1000$.

T.I.7.014

- (a) 8.83 — *Sum* $53 \div 6$.
(b) 11 — *max–min*.
(c) 13 — $5 \times \text{mean}$ — *sum of four*.

T.I.7.015

- (a) 13:25 — *Add 12 for pm (except 12 pm)*.
(b) 3:00 — *Add minutes; roll hours*.
(c) 95 minutes — *Arrival – departure*.

T.I.7.016

- (a) 19:05 — *Add 12 for pm (except 12 pm)*.
(b) 7:30 — *Add minutes; roll hours*.
(c) 25 minutes — *Arrival – departure*.

T.I.7.017

- (a) 22:50 — *Add 12 for pm (except 12 pm)*.
(b) 12:25 — *Add minutes; roll hours*.
(c) 95 minutes — *Arrival – departure*.

T.I.7.018

- (a) 160 mm — $\times 10$.
(b) 15 kg — *Convert to same unit*.
(c) 1.6 l — $\div 1000$.

T.I.7.019

- (a) 100 mm — $\times 10$.
(b) 9 kg — *Convert to same unit*.
(c) 1 l — $\div 1000$.

T.I.7.020

- (a) £2.09 — *Add pence; convert*.
(b) £2.91 — *Subtract*.
(c) Yes, £5.32 — *Compare sum with 1000p*.

T.I.7.021

- (a) £2.62 — *Add pence; convert*.
(b) £2.38 — *Subtract*.
(c) Yes, £5.54 — *Compare sum with 1000p*.

T.I.7.022

- (a) £2.75 — *Add pence; convert*.
(b) £2.25 — *Subtract*.
(c) Yes, £4.27 — *Compare sum with 1000p*.

T.I.7.023

- (a) £1.92 — *Add pence; convert*.
(b) £3.08 — *Subtract*.
(c) Yes, £6.55 — *Compare sum with 1000p*.

T.I.7.024

- (a) 22:50 — *Add 12 for pm (except 12 pm)*.
(b) 12:25 — *Add minutes; roll hours*.
(c) 95 minutes — *Arrival – departure*.

T.I.7.025

- (a) 19:55 — *Add 12 for pm (except 12 pm)*.
(b) 9:30 — *Add minutes; roll hours*.
(c) 95 minutes — *Arrival – departure*.

T.I.7.026

- (a) 22:30 — *Add 12 for pm (except 12 pm)*.
(b) 12:05 — *Add minutes; roll hours*.
(c) 95 minutes — *Arrival – departure*.